Title of your report: what you asked and what you found

Your Name

Data Analytics, Batch *N*

Partner: Name of the other student in your batch

November 2, 2026

Abstract

HINT (about 100 words): the outcome variable, the sample (how many cafés), the analyses you cover, and the two or three main results with their effect sizes in the original units. No new information that is not in the text below.

1. Introduction

HINT: two or three sentences on why the price (or rating, or popularity) of a café might depend on the variables of your batch. State the research questions in plain words. Do not describe the methods here.

2. Data and methods

2.1 Data

HINT: where the data come from (collected by students in the class), the number of cafés in the raw file and in the analysis dataset, every cleaning step your cleaning script applies (empty rows dropped, district labels harmonised, thousands separator removed, blank type set to missing, rule for cells with several prices) with the number of cafés affected, and the number of missing values in each variable you use. Say which variables were log-transformed and why. Keep this part short: the full descriptive statistics (Table 2) and the distribution checks are in Appendices A and B, and the main text only mentions what a reader needs in order to follow the results.

2.2 Statistical methods

HINT: one short paragraph. State $\alpha = .05$, the tests you used for each question, why you chose them (look at the plot, the sample size and the shape of the distribution), how confidence intervals were computed (including the bootstrap settings and the seed), which adjustment was applied to multiple comparisons, and the software with version numbers.

3. Results

HINT: the main text carries only the main results; checks, diagnostics and robustness analyses go to the appendices and are cited from here. Start with one captioned summary table of the main results of all ten analyses (`tables/summary.tex`), then load each analysis' table and figure by its name, each inside a table or figure environment with a `\caption` and a `\label` (for example `tables/A2_table.tex` and `figures/A2_figure.pdf`). One subsection per analysis. In

each, in this order: the question and the hypotheses with the direction fixed in advance; the plot; the estimate with its confidence interval; the effect size with its confidence interval; the test statistic with degrees of freedom and the exact p-value; one sentence of interpretation in the units of the data. Use the macros from the style file so that the format is identical everywhere, for example:

`\meanci{XX.X}{XX.X}{XX.X}{CZK}` gives M = XX.X CZK, 95% CI [XX.X, XX.X]
`\diffci{X.X}{X.X}{X.X}{CZK}{0.XX}{0.XX}{0.XX}` gives M_Δ = X.X CZK, 95% CI [X.X, X.X], d = 0.XX, 95% CI [0.XX, 0.XX]
`\tstat{XX.X}{X.XX}, \pval{.XXX}` gives t(XX.X) = X.XX, p = .XXX
`\Fstat{X}{XX.X}{X.XX}` gives F(X, XX.X) = X.XX

3.1 A1: One-sample test

Question, hypotheses, plot, estimate, interpretation. Remember that the benchmark was fixed before looking at the data.

3.2 A2: Two groups

State the direction of the subtraction in words. Report the mean difference with its confidence interval and Cohen's d with its confidence interval.

3.3 A3: Paired comparison

Report the mean of the within-café differences, its confidence interval and d_z. Show the differences in a plot.

3.4 A4: Three or more groups

Report the omnibus test, η^2, and the post hoc comparisons with adjusted p-values and confidence intervals. Show group means with confidence intervals in Figure 1.

Missing figure file: `figures/A4_figure.pdf`

Figure 1. Self-contained caption: what is on the axes (with units), what the points and bars show, what the error bars are (95% confidence intervals), and the sample sizes.

3.5 A5: Association of two categorical variables

Report the counts and percentages, the test with N, the smallest expected count, Cramér's V with its interval, and the cells with the largest standardised residuals.

3.6 A6: Correlation

Report r with its confidence interval, n, and the rank-based coefficients next to it. Say nothing about causes.

3.7 A7: Simple regression

Report the slope in the units of the outcome per unit of the predictor, its robust confidence interval, R^2 and what the residual plots show.

3.8 A8: Multiple regression

Report both models side by side in Table 1, the nested F-test and the change in the slope of interest. Mention the variance inflation factors.

Table 1. Regression estimates with robust standard errors in parentheses and 95% confidence intervals. The outcome is stated in the table note.

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3.9 A9: Interaction

Report the interaction coefficient with its confidence interval, the nested F-test, the slope of interest in each group with its interval, and show the fitted lines by group over the data.

3.10 A10: Logistic regression

State how the binary outcome was created (the median and its value, or the definition of the indicator). Report the average marginal effect of distance in percentage points with its 95% confidence interval as the headline, the odds ratio only as a check, the number of observations and events, and show a predicted-probability plot.

4. Discussion

HINT: what the results together say about your outcome, which results are exploratory, and what a reader should and should not conclude. Discuss at least these limits: a convenience sample collected by students, observational data, and prices of cafés in the same district that are not independent of one another. Write "in this sample" unless you can defend a wider claim.

A. Descriptive statistics

HINT: the full description of every variable used (the output of your descriptive script): n, number missing, mean with 95% confidence interval, standard deviation, median, interquartile range and range, and counts and percentages for categorical variables.

Table 2. Descriptive statistics of the variables used. Means are shown with 95% confidence intervals.

Missing table file: `tables/descriptives.tex`

B. Distributions and normality

HINT: what the plots and checks of your descriptive script showed for the variables you test: skewness, outliers, the shape within each group, and the result of the normality check. State which test you treat as primary for each question as a consequence.

Missing figure file: `figures/distributions.pdf`

Figure 2. Self-contained caption: the variables shown, the plot type (histogram, density, boxplot, Q–Q plot), and the sample sizes.

C. Regression diagnostics

HINT: residual plots, Q–Q plots of the residuals, leverage and influential observations, variance inflation factors, and whether the conclusion survives without the influential observations.

D. Alternative tests and robustness

HINT: the second method named for each question (parametric or rank-based), alternative specifications, bootstrap results, the full post hoc tables, and anything else that checks the main result.

E. Reproducibility

HINT: the repository link and the names of your three scripts (cleaning, descriptives, tests), the R version, the packages used, and the seed for all bootstrap procedures.